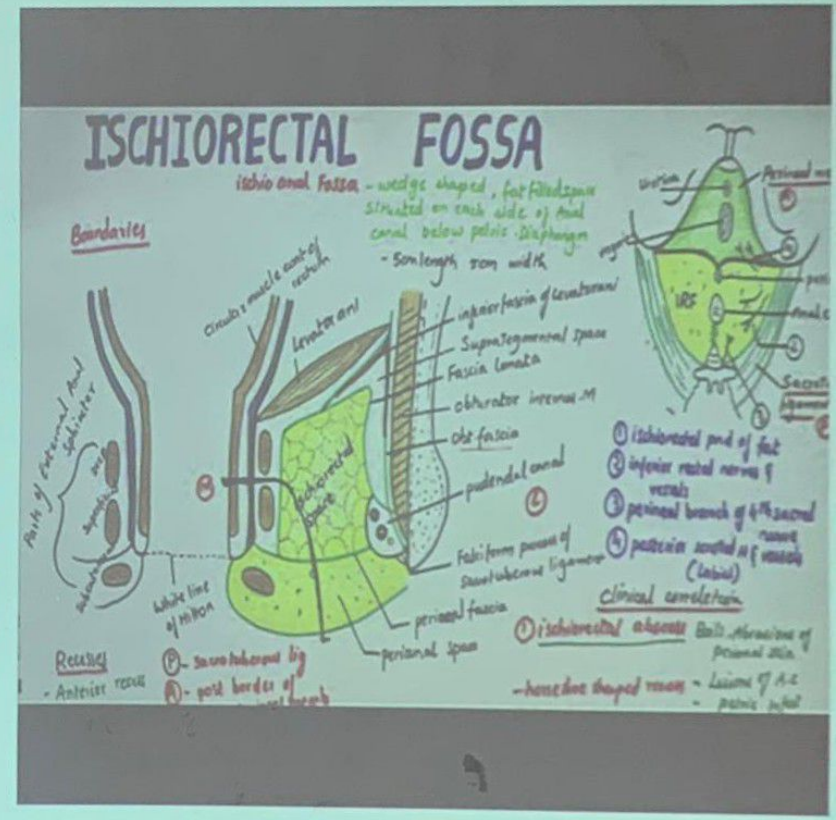
Ischio-rectal fossa

- Introduction
- Shape and location
- Boundaries
- Content
- Applied Anatomy
- Alcock canal

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Ischio-rectal fossa (ischio-anal fossa): It is a pair of wedge-shaped space predominantly filled with adipose tissue, located on either side of the anal canal, below the pelvic diaphragm. It acts as a cushion for neurovascular structures and allows for anal canal expansion during defecation.

It is situated in such a way that the base is posterior inferior, while the apex is anterior superior, located between obturator internus lateral and levator ani medial.

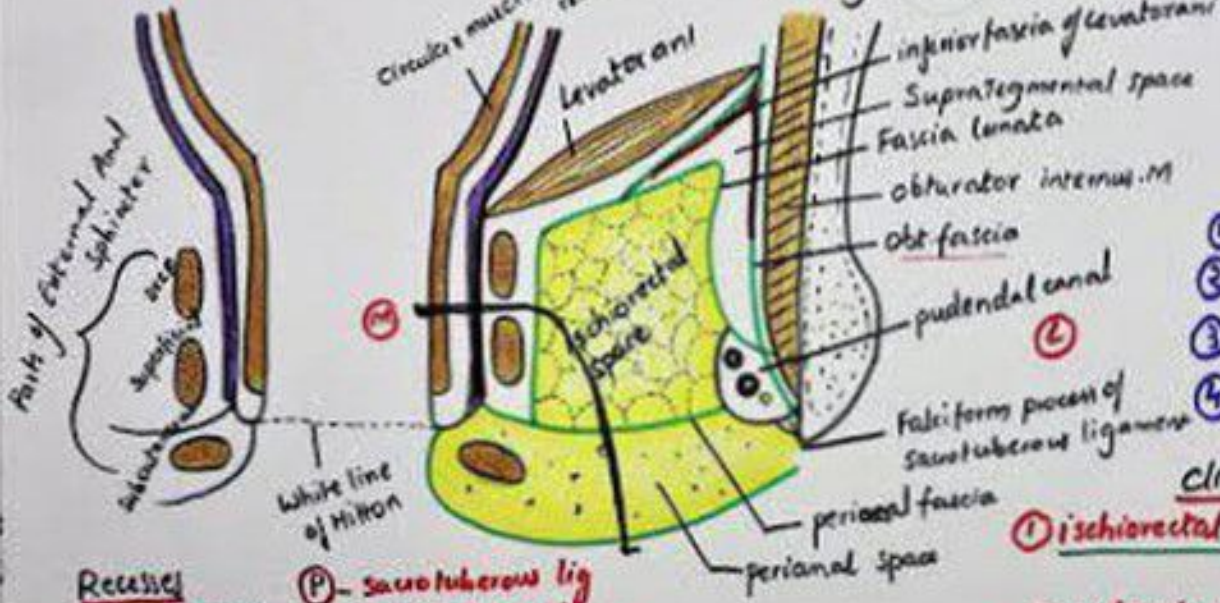


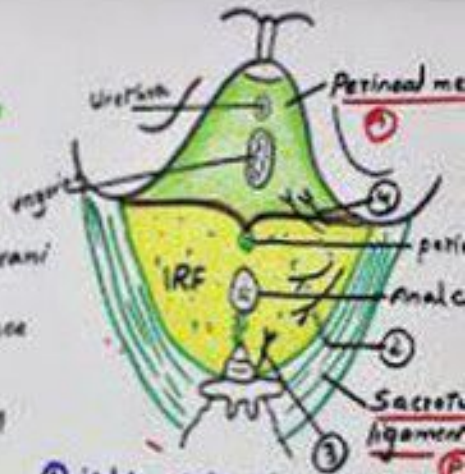
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ISCHIORECTAL FOSSA

ischio anal fossa - wedge shaped, fat filled space situated on each side of anal canal below pelvic diaphragm
- 5cm length, 3cm width

Boundaries





1. ischiorectal pad of fat
2. inferior rectal nerves & vessels
3. perineal branch of 4th sacral nerve
4. posterior scrotal N & vessels (Labial)

Recesses

- Anterior recess
- (P) - sacrotuberous lig
- (A) - post border of

Clinical correlation

1. ischiorectal abscess - Boils, Abrasions of perineal skin
   - horseshoe shaped recess
   - Lesions of A.C
   - pelvic infect

It has the following boundaries:

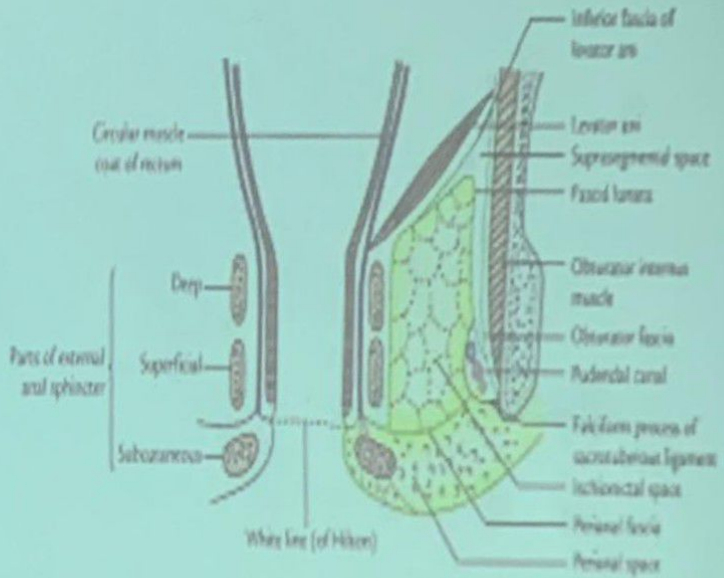
Base: By skin

Apex: Is formed where the obturator internus fascia meet inferior fascia of pelvic diaphragm.

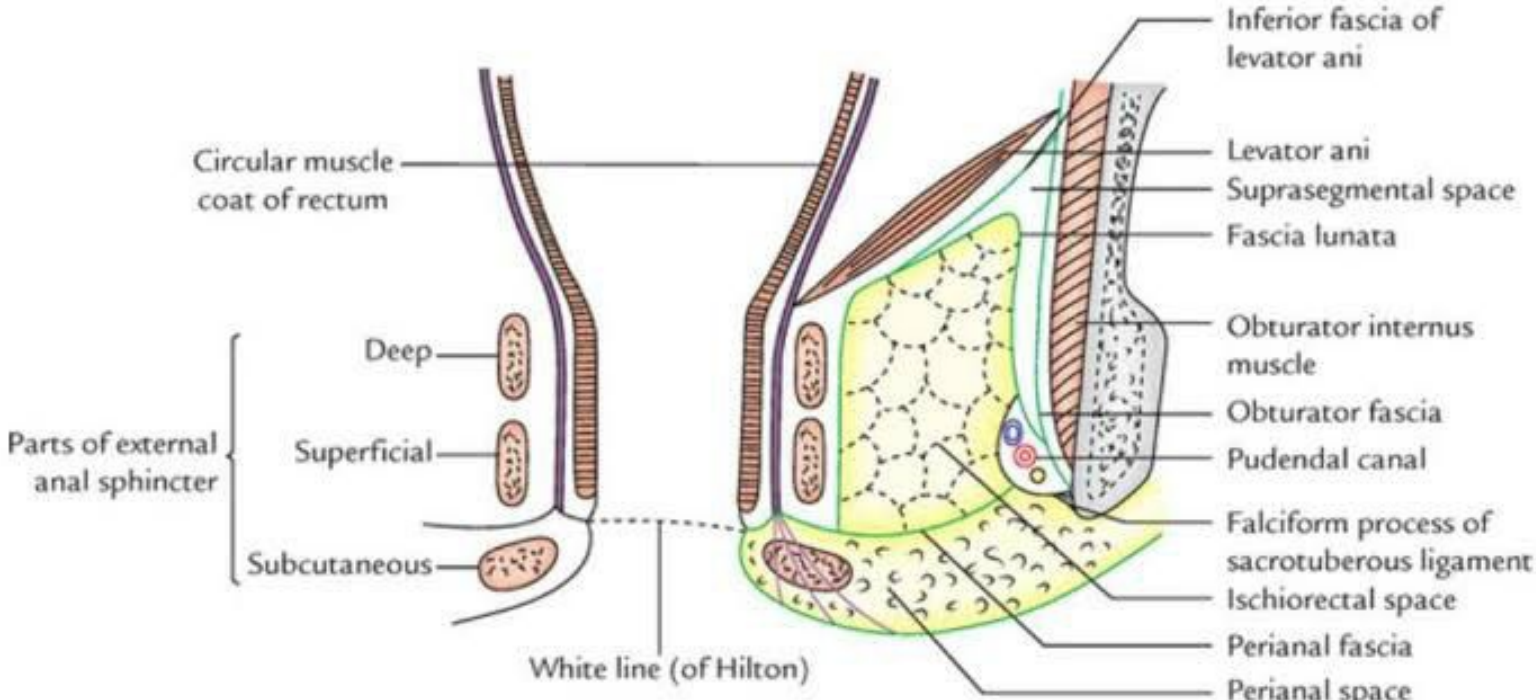
Lateral: Obturator internus muscle and fascia

Medial: External anal sphincter and levator ani muscles.

Anteriorly: relates to the posterior margin of perineal membrane, comprising, superficial and deep transverse perineal muscles



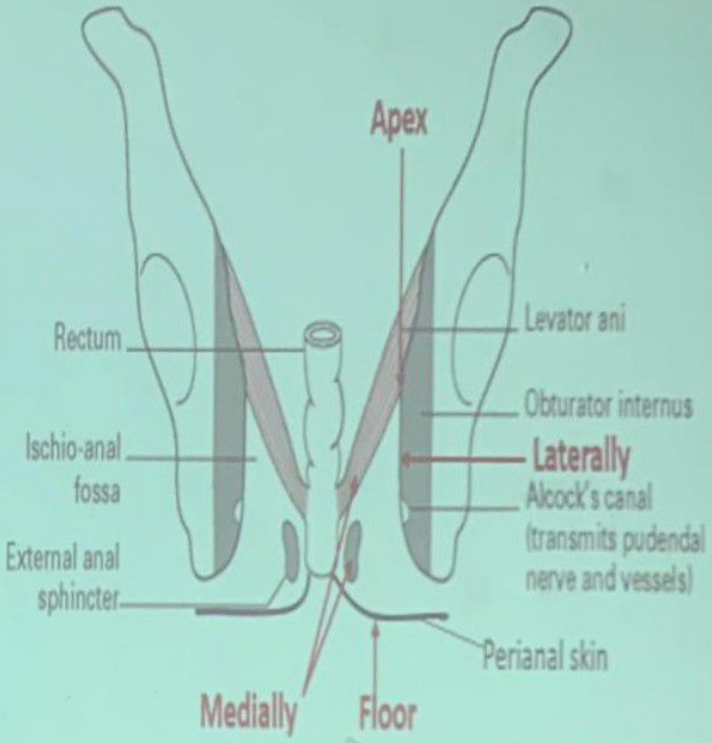
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Inferior fascia of levator ani
Circular muscle coat of rectum
Levator ani
Suprasegmental space
Fascia lunata
Deep
Parts of external anal sphincter
Superficial
Obturator internus muscle
Obturator fascia
Pudendal canal
Falciform process of sacrotuberous ligament
Subcutaneous
Ischiorectal space
Perianal fascia
White line (of Hilton)
Perianal space

Posteriorly: Gluteus maximus muscle and sacrotuberous ligament.

Content: Filled with adipose tissue (pads of fat), Pudendal nerve, Internal pudendal vessels, and Peroneal branches of S3 and S4.

Clinical Relevance: The fossa is a common site for abscesses (ischiorectal abscesses) due to its proximity to the anal canal, often presenting with pain or swelling. It can also be a site for tumours, which are often benign but may require MRI imaging for assessment.



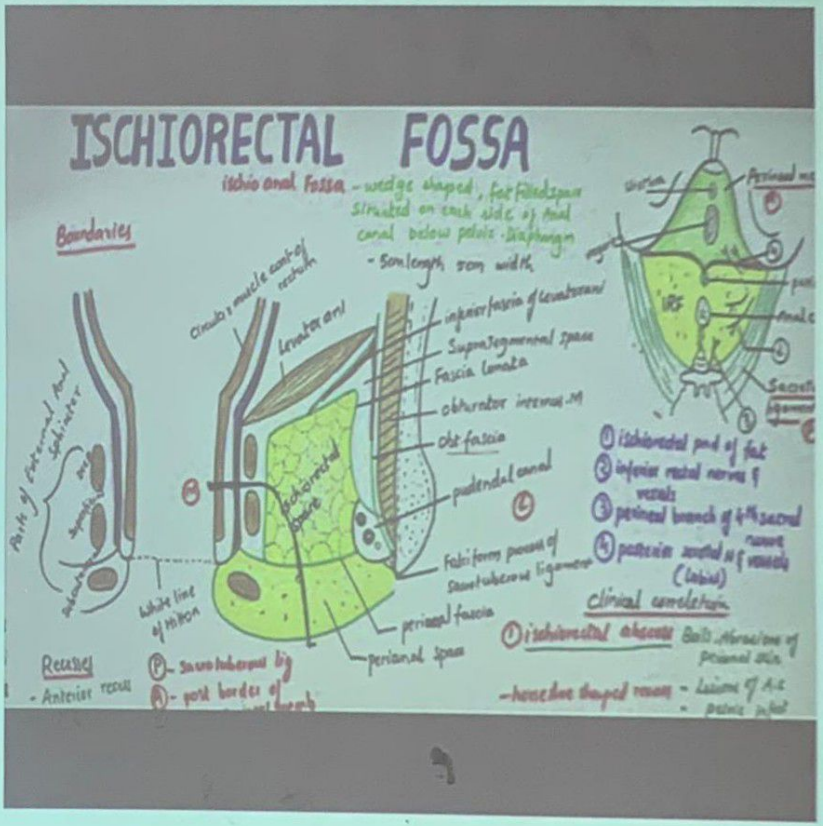
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Pudendal canal: (Alcock's canal) is a fascial sheath in the lateral wall of the ischio-anal fossa that protects the pudendal nerve, internal pudendal artery, and vein.

Location: Situated on the obturator internus muscle, in the lower lateral wall of the ischio-rectal fossa.

Structure: Formed by the obturator fascia, extending from the lesser sciatic notch to the perineal membrane.

Contents: It contains the pudendal nerve, internal pudendal artery and internal pudendal vein.



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Clinical Significance

Pudendal Neuralgia: Inflammation or compression of the pudendal nerve within this canal can cause chronic pain (pelvic pain, burning sensations), often worsened by sitting.

Pudendal Nerve Block: A medical procedure targeting this area, typically for pain management, chronic pain treatment, or during childbirth.

A pudendal nerve block is a regional anaesthesia procedure that injects local anaesthetic, sometimes with steroids, around the pudendal nerve to relieve pain in the perineum, anus, and genitalia.

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